

Chapter 3: Complex-Plane Readout Figures — Three Grids: Step 0, Final Step, and Zoom into the Condensed Center

Noriaki Kihara (WF System Co., Ltd.) ORCID 0009-0004-6753-4020

September 5, 2026 Version DOI 10.5281/zenodo.22317636

(Chapter 3 of the reproduction papers for the $N=3..40$ stage-1+2+3 sweep. Shares the Concept DOI 10.5281/zenodo.22317635 with the overview paper; Version DOI 10.5281/zenodo.22317636. Equation numbers continue from Chapters 1–2 (Eqs. 1–25). Code blocks are quoted verbatim; comments inside them remain in Japanese, as in the originals.)

1. Purpose

This chapter describes the process that generates, from the state npz files saved by the Chapter-2 sweep (the $\Delta \tau = 2 \pi / N$ runs), three grid figures reading out the per-edge complex waves $z_e \in \mathbb{C}$ on the complex plane — (1) step 0, (2) the final step (step 500), and (3) a zoom into the largest angular cluster at the final step. The design intent of the three figures (their division of roles corroborating the start, end, and end-interior of the inflation figure from the configuration side, the caution that “the complex plane of the figures $\neq$ the parent plane Π ”, and the bridging identity) is given in **Chapter 2, §2.6 (Eq. 25)**; this chapter fixes the implementation, execution, and observations. The data are read-only; the dynamics and stored data are never touched.

2. Theoretical Background

The mathematics of states and measurement follows Chapter 2 (Eqs. 22–25). The figure-specific conventions of this chapter are defined as Eqs. 26–28.

(Eq. 26) Duplicate counting (notation of degeneracy) — The set of plotted complex values $\{z_e\}$ is partitioned into equivalence classes by **rounding the real and imaginary parts to the 12th decimal place**; classes with $c > 1$ members are annotated “xc” at their position:

```
class(w) = ( round(Re w, 12), round(Im w, 12) )
```

Rounding at 12 digits is coarser than the effective double precision (~ 16 digits): it identifies differences at the level of rounding error while keeping physically significant separations (relative 10^{-4} or larger in this series) distinct. In the zoom figure the rounding is tightened to the 15th decimal (Eq. 28), so that only machine-precision coincidences are counted as “xn”.

(Eq. 27) Panel scaling convention — The axis range of each panel is the square $[-1.15r, +1.15r]$ (equal aspect) with $r = \max_e |z_e|$ the overall amplitude of that panel, and **the tick values display**

the actual values (no relabeling by normalized values). The line segments from the origin to each point are guides for reading the argument and modulus of each wave simultaneously.

(Eq. 28) Extraction of the largest angular cluster (algorithm of the zoom figure) — The final-step state, in coordinates normalized by the overall amplitude, is grouped by rounding at **coarse resolution 1/100**; the group with the most members is the zoom target:

```
key(w) = ( round(Re w / amp, 2), round(Im w / amp, 2) ),    amp = max_e |z_e|
C* = argmax_{key} |{ w : key(w) = key }|
center c = mean(C*),    spread = max_{w ∈ C*} |w - c|,    window = 1.4 × spread
```

The panel title prints $|C^*|$, spread, and amp. The spread quantifies whether the cluster is an exact single point (exact degeneracy) or a bundle of finite width (the zoom row of the table in Chapter 2, §2.6).

3. Implementation Method

- Plotting program `plot_complex_plane_N3_N40_stage123_v1.py` (bundled in this package; read-only). The drawing style extends the existing plotting programs of this series (the grid style of `complex_plane_readout_step0_step2000_20260904/plot_complex_plane_step0_step2000.py` and the cluster-zoom algorithm of `自発的分裂予備実験_v1/N40_state_readout_20260904/plot_complex_plane_N40` to the 8×5 grid of $N=3..40$; the algorithms themselves are identical (the style lineage is unchanged).
- The input is solely the Chapter-2 product `results/hm_N{N}_den_{N}_states_500.npz`. **The den=N ($\Delta \tau = 2 \pi / N$) run is used as the representative for each N** (step 0 is identical across denominators, so the representative choice matters only for the final step; final configurations for other denominators can be read out likewise from the stored npz).
- The output is 3 PNG files. No data are rewritten or regenerated.

4. Detailed Design

4.1 Overall Flow

```
[initialization]  fix the input folder (results/)
[loop]   draw_grid(step=0)  : draw the step-0 state on each panel for N=3..40 (Eqs. 26, 27)
        draw_grid(step=500): same for the final-step state
        zoom grid          : largest angular cluster (Eq. 28) per panel for N=3..40
[finalization]  savefig each figure; switch off the 2 unused panels
```

4.2 Overall Data Flow

- **Input:** `results/hm_N{N}_den_{N}_states_500.npz` $\times 38$ (Chapter-2 products; read-only). Fields used: `Z` ($501 \times M$), `denominator`, `steps` (for consistency asserts)
- **Parameters** (all constants inside the program):
 - rounding digits: 12 (Eq. 26) / 15 (inside the zoom, Eq. 28) / coarse resolution 2 digits (the 1/100 of Eq. 28)
 - scale factor 1.15 (Eq. 27), window factor 1.4 (Eq. 28)
 - grid 8×5 (38 used, 2 off), dpi=180
- **Output:**

- fig_complex_plane_step0_N3_N40_stage123.png
- fig_complex_plane_final_N3_N40_stage123.png
- fig_complex_plane_final_zoom_N3_N40_stage123.png

4.3 Individual Processes

■4.3.1 Initialization (loading and consistency checks)

```

18 def load(N, step):
19     d = np.load(os.path.join(IN, f'hm_{N}_{N}_den_{N}_states_500.npz'))
20     assert int(d['denominator']) == N and int(d['steps']) == 500
21     return np.asarray(d['Z'][step], dtype=np.complex128)

```

- The assert on line 20 checks that “the intended file (den=N, 500 steps) is being read”; on mismatch an AssertionError stops execution (no drawing occurs).

■4.3.2 Loop (1): Grid Rendering (Eqs. 26, 27)

```

26     for k, N in enumerate(range(3, 41)):
27         ax = axs[k]
28         z = load(N, step)
29         M = N * (N - 1) // 2
30         assert z.size == M
31         for w in z:
32             ax.plot([0.0, w.real], [0.0, w.imag], color='tab:blue', linewidth=0.7, alpha=0.6)
33             ax.plot(z.real, z.imag, 'o', ms=2.5, color='tab:red', alpha=0.85, linestyle='none')
34             cnt = Counter((round(float(w.real), 12), round(float(w.imag), 12)) for w in z)
35             for (a, b), c in cnt.items():
36                 if c > 1:
37                     ax.annotate(f'x{c}', (a, b), textcoords='offset points', xytext=(3, 3),
38                               fontsize=5, color='black')
39             r = float(np.abs(z).max())
40             lim = r * 1.15 if r > 0 else 1.0
41             ax.set_xlim(-lim, lim); ax.set_ylim(-lim, lim); ax.set_aspect('equal')

```

- Lines 31–32: segments from the origin; line 33: points; lines 34–38: the duplicate annotations of Eq. 26; lines 39–41: the scaling convention of Eq. 27 (tick values remain actual values; ticklabel_format on line 46 only switches the notation to scientific).

■4.3.3 Loop (2): Zoom into the Largest Angular Cluster (Eq. 28)

```

67     z = load(N, 500)
68     amp = float(np.abs(z).max())
69     coarse = {}
70     for w in z:
71         key = (round(float(w.real) / amp, 2), round(float(w.imag) / amp, 2))
72         coarse.setdefault(key, []).append(w)

```

```

73     mem = max(coarse.values(), key=len)
74     zz = np.array(mem)
75     c = zz.mean()
76     dev = np.abs(zz - c)
77     spread = float(dev.max())
78     win = spread * 1.4 if spread > 0 else amp * 1e-12
...
90     ax.set_title(f'N={N}: {len(zz)} waves, dev={spread:.2e} (|z|max={amp:.2e})', fontsize=7)

```

- Lines 70–73: the grouping and largest-cluster selection of Eq. 28; lines 75–78: center, spread, window; line 90: printing member count, spread, and amp in the panel title (this printed output is the source of the observed values in §6).

■4.3.4 Finalization and Exceptional Behavior (elements not formulated as equations)

- Line 40, `lim = r*1.15 if r > 0 else 1.0`: fallback for the degenerate case of all points at the origin (does not occur in these data).
- Line 78, `win = spread*1.4 if spread > 0 else amp*1e-12`: window fallback when the cluster has a single member (`spread=0`; occurs at $N=3$, where the largest cluster is a single wave).
- Lines 50–51 and 91–92: of the $8 \times 5 = 40$ panels, the unused 39th and 40th are set to `axis('off')`.
- The only exception paths are the asserts (lines 20, 30); there are no other branches.

5. Execution Results

5.1 Reproduction Commands

```

cd N3_N40_stage123_sweep_20260905
python3 plot_complex_plane_N3_N40_stage123_v1.py
# or as the final step of ./run_all.sh

```

5.2 Execution Environment

- Python 3.9.6 (`.venv/bin/python3`), numpy 2.0.2, matplotlib (Agg rendering)
- macOS 26.3.1 (arm64)

5.3 Execution Time

Under about 1 minute for the three figures in total (dominated by 38×2 npz loads and the rendering of up to 780 segments $\times$ 38 panels).

5.4 Verification Gates

Gate	Pass condition	Measured	Verdict
G1: input consistency	asserts denominator==N, steps==500, z.size==M pass at every load	passed for all 38 $N \times 3$ figures (no AssertionError)	PASS
G2: completion	ALL DONE printed, 3 PNGs generated	confirmed	PASS

(The bit-level lineage of the input npz themselves is guaranteed by Chapter 2 G1 — Z[0] of all 228 runs bit-identical to the static parents.)

5.5 Data

Item	Content
Input	<code>results/hm_N{N}_den_{N}_states_500.npz</code> × 38 (Chapter-2 products; read-only)
Output location	package root
step-0 figure	<code>fig_complex_plane_step0_N3_N40_stage123.png</code> (632,949 bytes)
final-step figure	<code>fig_complex_plane_final_N3_N40_stage123.png</code> (4,582,367 bytes)
zoom figure	<code>fig_complex_plane_final_zoom_N3_N40_stage123.png</code> (494,906 bytes)
SHA256	the bundled <code>SHA256SUMS.txt</code> is canonical

5.6 Figures

The three figures above (8×5 grids; each panel equal aspect, actual-value ticks). How to read them, and their correspondence with the inflation figure, follow the table in Chapter 2, §2.6.

6. Execution Analysis (objective report and observations only; the numbers

originate from the in-figure printing = the output of this program)

1. **Step-0 figure:** every panel for $N=3..40$ shows the star shape of two antipodal pairs (4 bundles). The bundles have a radial spread of amplitudes. The duplicate annotations of Eq. 26 appear only for some small N (x4 at $N=4$; x2–x3 for $N=5–9$, etc.); exact duplicates vanish as N grows.
2. **Final-step figure:** in every panel the star has disappeared, replaced by a nearly equimodular ring-like arrangement (sparse spokes for small N). The Eq. 26 (12-digit) duplicate annotations do not appear except for a few small- N cases (x2 at $N=4, 5$).
3. **Zoom figure:** the largest angular cluster has 1–10 members ($N=3$: 1 wave; $N=39, 40$: 10 waves); the spread is of order $10^{-7-10}-4$ for $N \geq 6$ (e.g., at $N=40$, $\text{dev}=2.04\text{e-}04$ with $|z|_{\text{max}}=3.64\text{e-}02$,

relative $\sim 5.6 \times 10^{-3}$). Only the two-wave clusters of $N=4, 5$ coincide at machine precision (dev $10^{-16-10-10}$).

4. Summary of observations (facts only): for all $N \geq 6$ the final-step angular clusters are bundles of finite width, not condensation onto exactly identical complex values. As per the division of roles in Chapter 2 §2.6, this records the fine organization of the final state that $H \perp / H$ does not measure.

(End of Chapter 3. The overview paper is separate.)